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Executive Summary

This report presents a comprehensive analysis of the sales performance of an Australian chocolate company from 2022 to 2024, using a star schema data model in Power BI. The dataset covers six countries across four continents and includes detailed information on products, customers, and employees.

Key findings reveal a positive overall performance trend, with both revenue and profit showing consistent growth during the period. Australia remains the strongest market, while New Zealand underperforms despite its geographic proximity. From a product perspective, Bars and Premium categories drive the majority of revenue but deliver the lowest profit margins, whereas Truffles stand out as the most profitable category. Brand performance is led by ChocoLux and CocoaCraft, while SweetPeak continues to lag.

Customer analysis shows a well-distributed base, with Oceania and the Americas accounting for the largest share. Retail and Wholesale segments generate similar sales volumes, but Wholesale delivers slightly higher profits. A specific analysis also identified customers with consistently declining purchase patterns between 2022 and 2024.

Strategic Recommendations include focusing on margin improvement for high-volume products, expanding customer acquisition efforts in high-potential markets such as India and the USA, and developing tailored strategies per country to address differing product preferences.

This analysis provides actionable insights to support data-driven decision-making and highlights opportunities for sustainable growth.

Presentation

This report analyses the sales performance of an Australian chocolate company from 2022 to 2024. The objective is to deliver actionable insights into the company's markets, products, and customers.

To support this analysis, a star schema data model was built in Power BI.

The model consists of 6 tables: the fact table (*FactSales*), and 5 dimension tables (*DimProduct*, *DimGeography*, *DimEmployee*, *DimDate*, *DimCustomer*). All the dimension tables are connected to the fact table, as per shown in the figure below, with the columns of each table included:

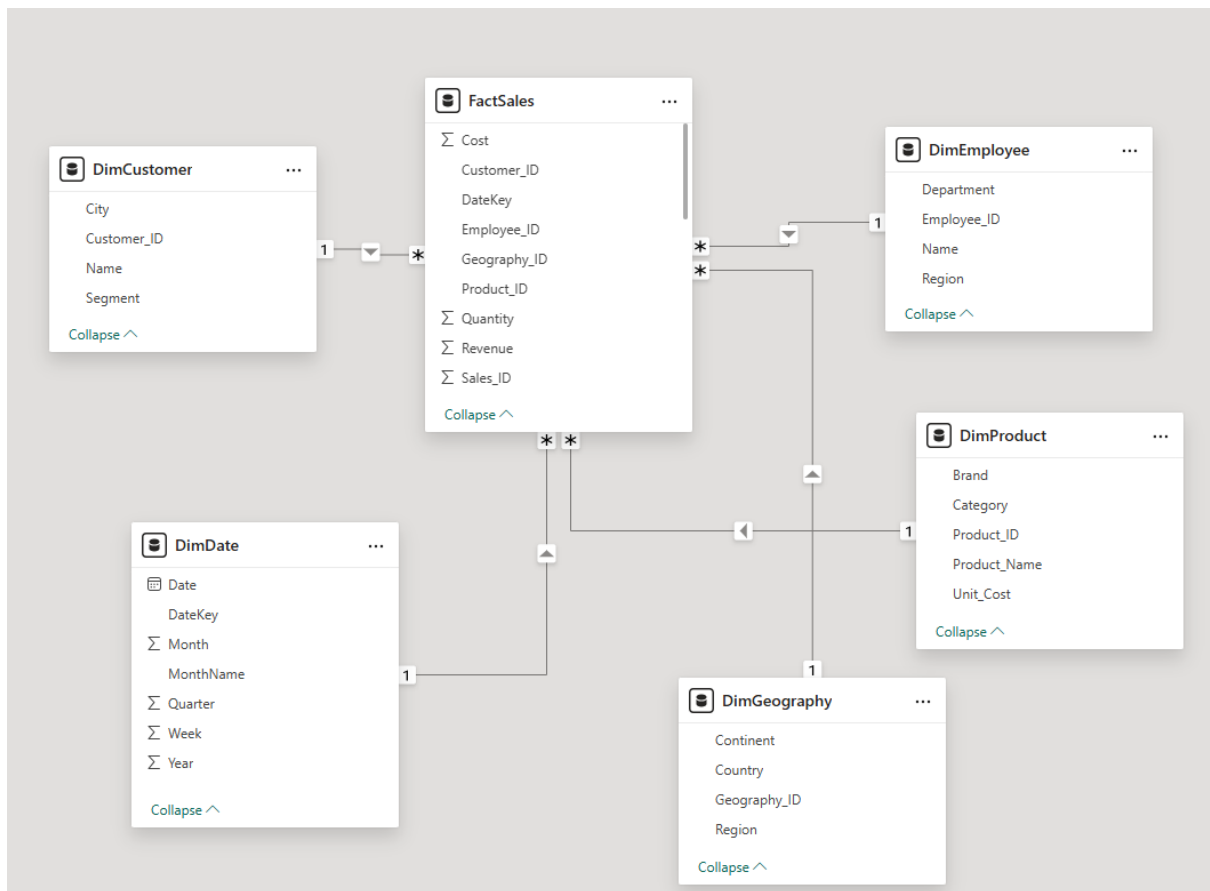


Figure 1 - Relational Model

This representation is referred to as a star schema, since all the dimension tables are only connected with the fact table, through the primary key – foreign key relation.

The sales in this database are from the years 2022 to 2024, with months ranging only from January to August, in 6 countries - Australia, Canada, India, New Zealand, UK and USA - from 4 different continents.

From the products, the database refers 4 categories (Bars, Premium, Truffles and Gift Boxes) and 4 brands (ChocoLux, CocoaCraft, Velvet Cocoa and SweetPeak).

Regarding the customers, they belong to one of 3 segments (Distributor, Wholesale and Retail).

The analysis of this company's sales, would focus on three parts, each one having a specific dashboard in the PowerBI file:

- ➔ Analysis of the sales in general;
- ➔ Analysis of the sales with focus on the products;
- ➔ Analysis of the sales with focus on the customers;

Analysis of the sales in general

The analysis of the sales starts by a general overview, with focus on variables like revenue, profit and location.

Starting by a comparative analysis between revenues and costs, we can see from the image below that there's a positive correlation between both: the higher cost of the sale, corresponds also to a higher revenue.

This positive correlation confirms the company's expectations and validates observations from previous internal reports.

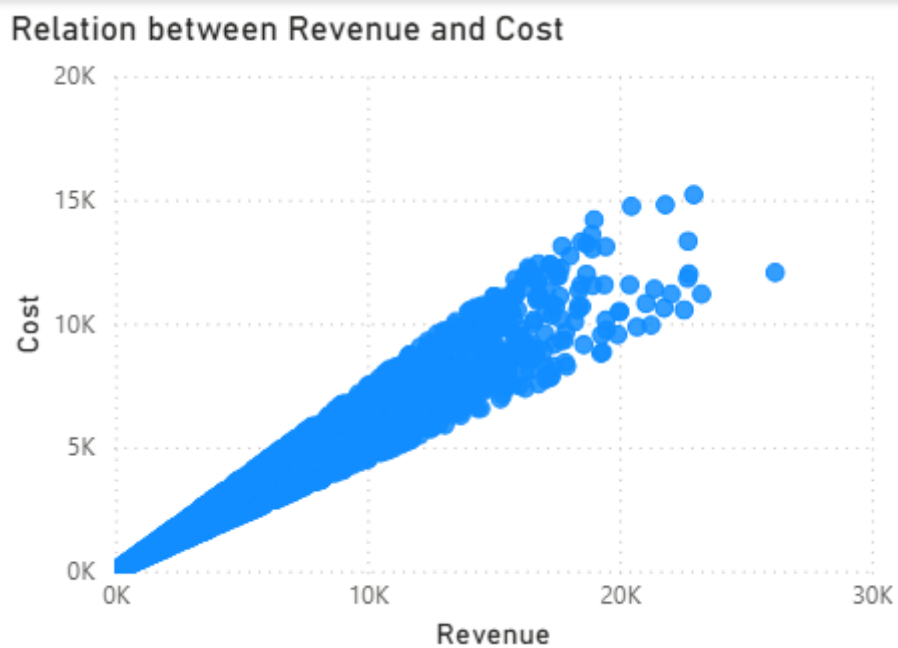


Figure 2 - Relation between Revenue and Cost

Analysing the evolution of revenue and profit through the time frame of our model it's possible to see that both show an increase.

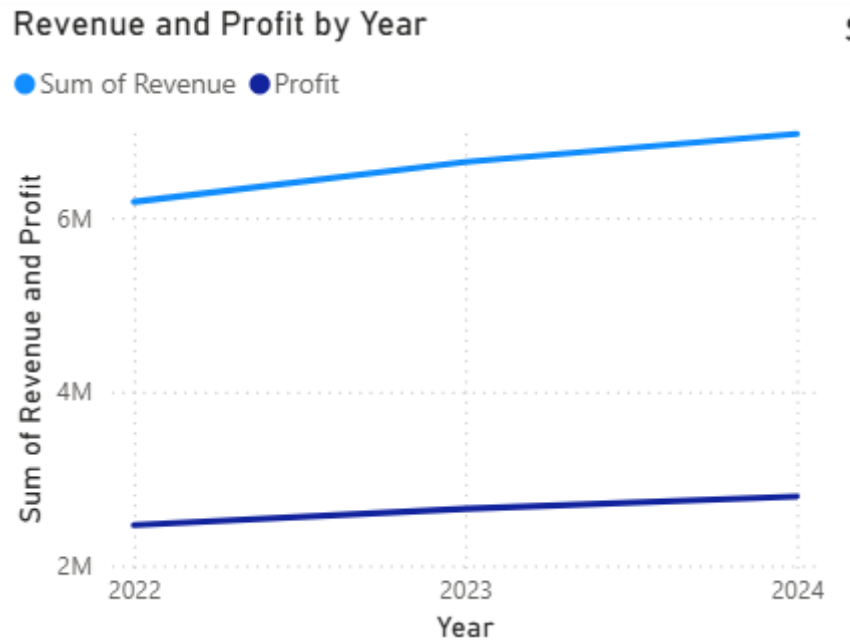


Figure 3 -Evolution of Revenue and Profit

This reflects the success of both the campaigns (to attract new customers) and the new policies to reduce costs by increasing efficiency in the production processes.

In all the analysis using the Profit, the values were obtained through a measure, that calculates the profit by subtracting the total costs from the total revenues.

Making a geographic analysis of the revenues, the figure below shows that the market that gives the biggest revenues is Australia, and the country that registered the lowest revenues is its neighbour, New Zealand.

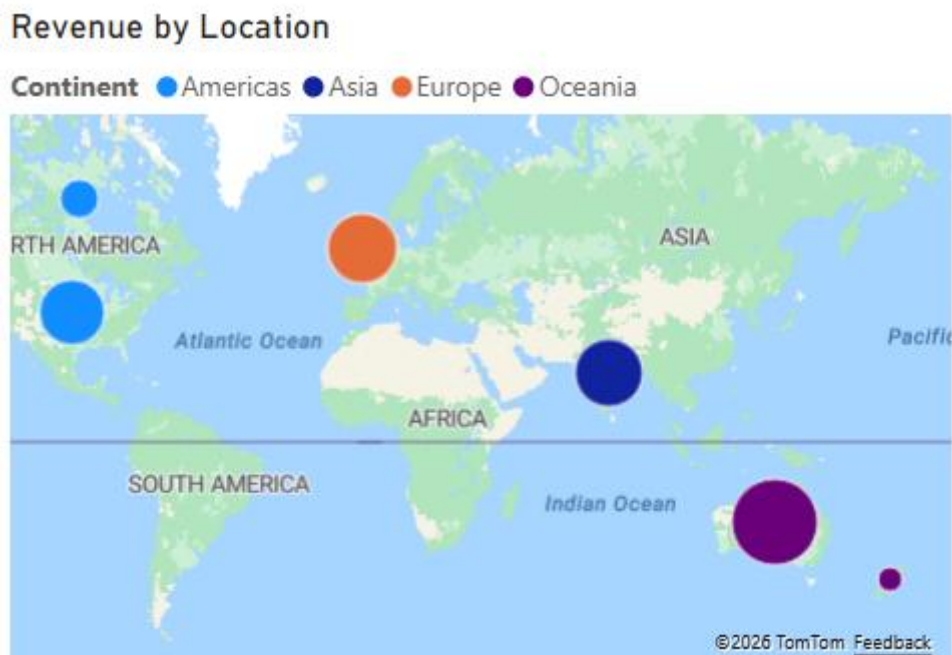


Figure 4 - Revenue of sales by Location

Being a company based in Australia, discovering that Australia was the biggest market in terms of revenue wasn't a surprise. However, the same cannot be said about the smallest market since, despite being a smaller market in terms of consumers, New Zealand is a neighbour and other competitors in the same sector have been thriving in the New Zealander market.

Focusing on the employees, the next figure shows the TOP10 best employees:

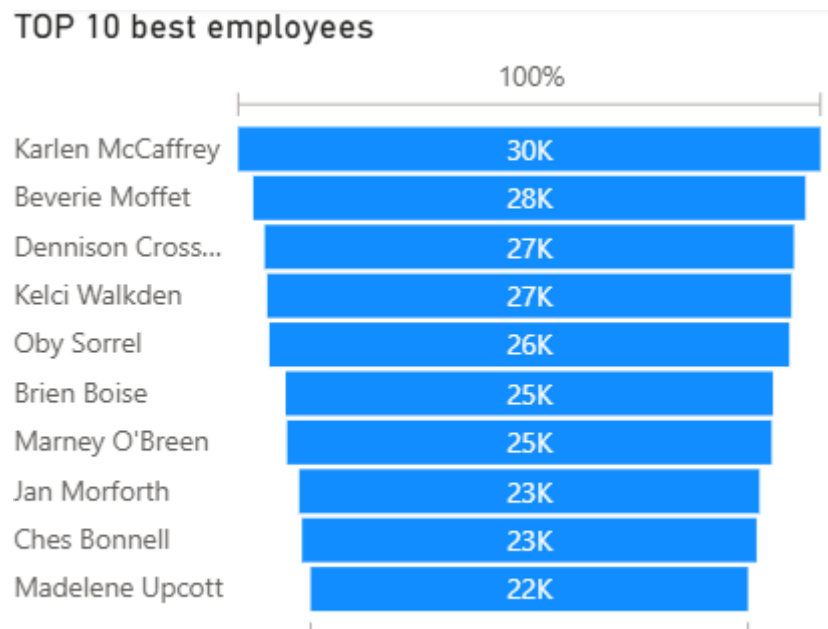


Figure 5 - TOP10 employees with highest sales

It's important to notice that the best employee sold around 30 thousand units, and the last in this TOP around 11 thousand.

Finally, the first page includes also decomposition tree that allows to make a Breakdown Analysis of profits by country, category and segment.

Analysis of the sales with focus on the products

Analysing the sales from the products perspective, we start by analysing the revenues by brand:

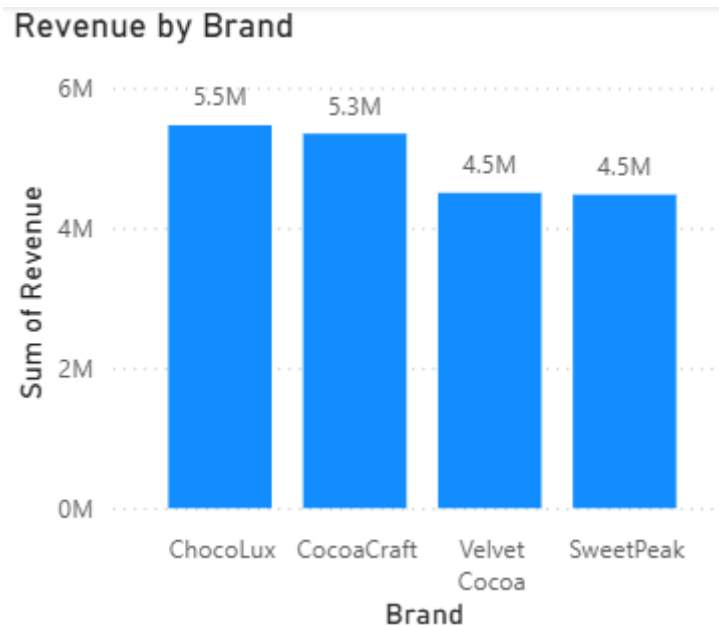


Figure 6 - Revenue of Sales by Brand

By analysing the column chart, it is easy to see that the brand with the highest revenue is "ChocoLux", immediately followed by "CocoaCraft", both with total revenues higher than 5 million Euros. The other two brands have the same total amount of revenues (4.5 million).

These results were largely expected. CocoaCraft, one of the company's oldest and most established brands, continues to perform strongly. However, despite the huge efforts to promote SweetPeak, the revenues are still disappointing.

A similar analysis by product category (see pie chart below) shows that Bars and Premium are the top performers, each contributing over 27% of total revenue. Truffles and Gift Boxes follow, with approximately 23% and 22% respectively.

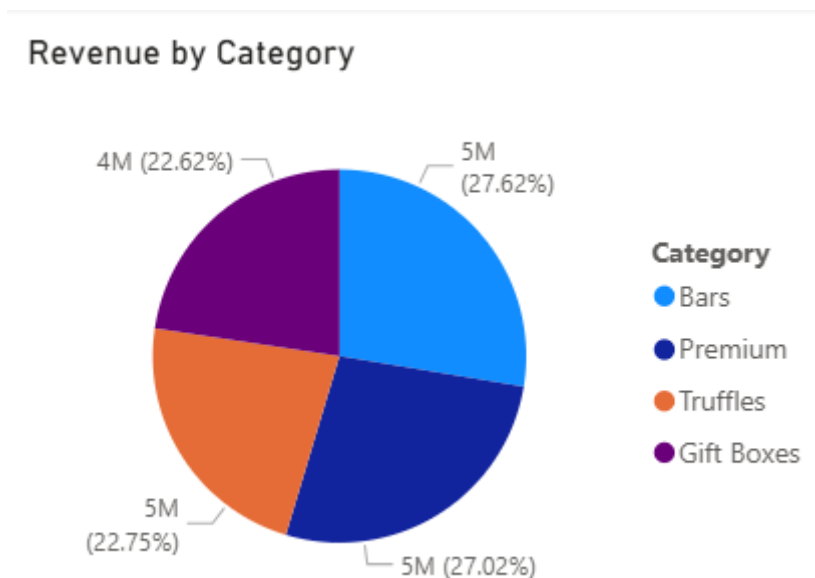


Figure 7 - Revenue of sales by Category

This analysis met what the company was expecting, and what was seen in the reports from the previous periods (before 2022).

Moving now to an analysis by product name, we can see on this following Treemap that, the product that have the highest profit to the company was Smooth Silky Salty (with more than 448 thousand Euros), followed by 50% Dark Bites and Peanut Butter Cubes, both with 430 thousand Euros.

The product that gave the lowest profit to the company was Spicy Special Slims, with less than 371 thousand Euros.

Profit by Product Name

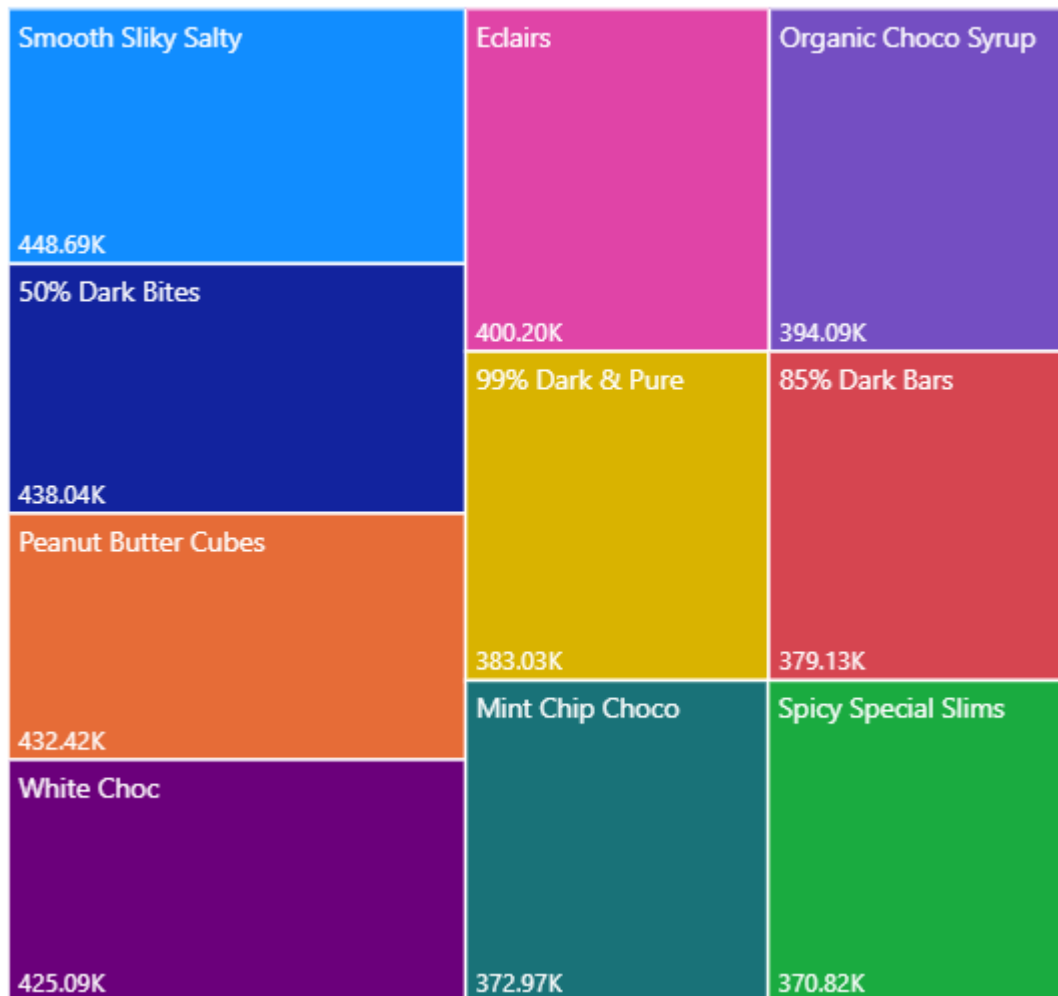


Figure 8 - Profit of Sales of each Product

If we analyse the most sold product by country we can conclude from the table below that, the most sold product was different in each country.

Most sold product by country

Country	Top Product by Country	Top Product Sales
Australia	50% Dark Bites	9692
New Zealand	Mint Chip Choco	7768
Canada	Choco Coated Almonds	7420
UK	Milk Bars	6903
USA	Fruit & Nut Bars	6276
India	Spicy Special Slims	6199

Figure 9 - Most sold product in each country

This highlights the importance of localized strategies, as customer preferences vary significantly across countries.

To make this table, it was necessary to create two measures. The first one, to obtain the Top product by country and the sales of that top product.

The last visualization is a scatter plot that shows the relation between profit margin and revenue.

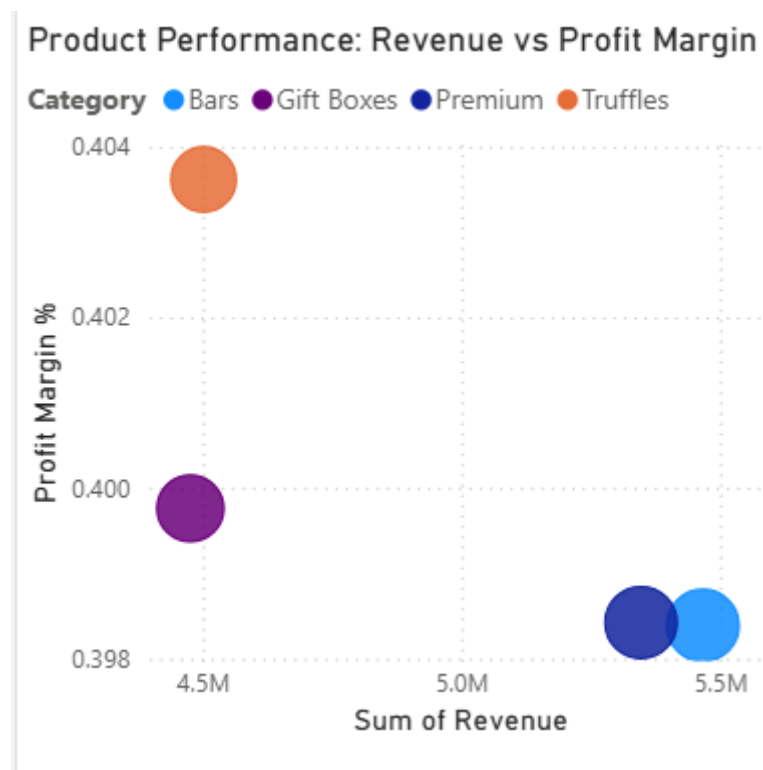


Figure 10 - Performance of each category

For the elaboration of this plot, a measure was created: the profit margin, consisting of the relation between total profit and revenue.

This chart allows us to conclude that the bars and Premium generate the highest revenue but have the lowest profit margins. This shows us that they are high-volume, low-margin products (they bring a lot of money but don't keep much profit after deducting the costs). As for the Truffles, they generate a lower revenue but have the highest profit margin among all the product categories. This category is where the company makes more profit per dollar sold, even if the total sales are lower.

Finally, regarding the gift boxes, they have a decent revenue and margin.

Analysis of the sales with focus on the customers

In the last chapter we make a more detailed analysis of the sales with a focus on the customers.

Starting by analysing the geographical distribution of the customers, we see a dominance of the Americas and Oceania, with this last taking the first place, with 87 customers, with 85 from the Americas.

Customer Continent	Total Customers by Continent
Americas	85
Asia	39
Europe	39
Oceania	87
Total	250

Figure 11 - Distribution of Customers by Continent

Asia and Europe have the smallest customer bases, with 39 customers each.

These values were obtained from a measure. To obtain this table, firstly was necessary to add a new column to the DimCustomer table, to indicate the continent of origin of each customer. Then a measure was calculated to sum the number of customers from each continent of origin, through the Customer_ID column.

Moving to the TOP10 of customers by revenue, we can see in the visualization below that the best customer in terms of revenue brought to the company is “Layla Green”, with around 411 thousand Euros, and the only one above 400 thousand Euros.

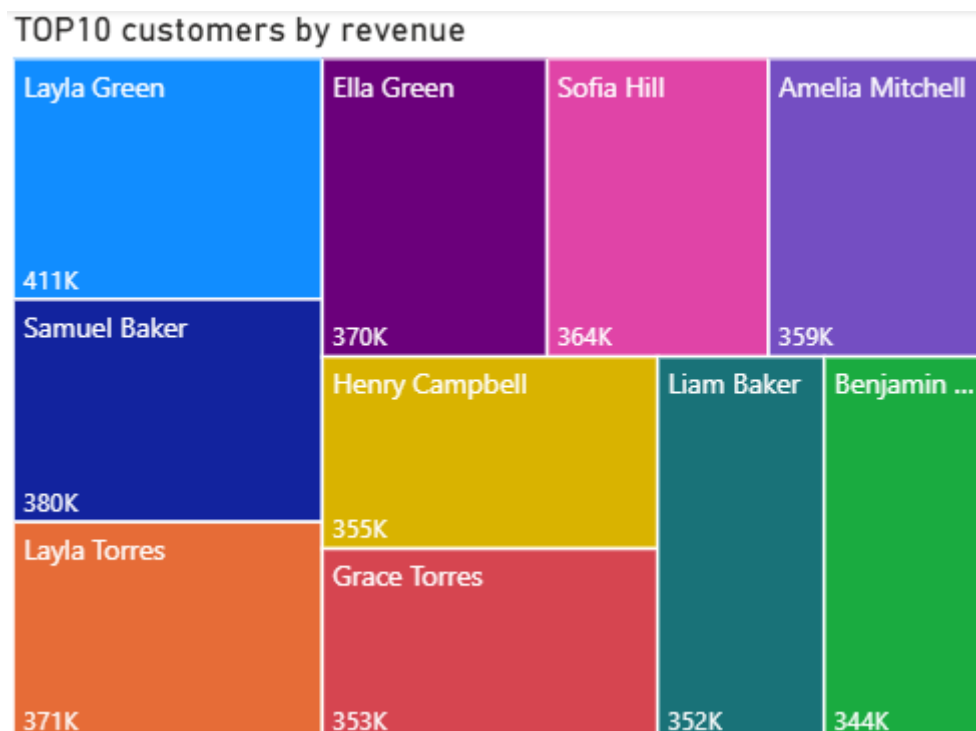


Figure 12 - TOP10 customers by revenue

The second customer – Samuel Baker – gave around 380 thousand Euros in revenues, and the third one 371 thousand (Layla Torres). It’s also important to highlight that all the customers in the TOP10 gave more than 340 thousand Euros in revenues to the company.

Segment analysis reveals that Retail purchased the highest volume of products (183 thousand units), closely followed by Wholesale (182 thousand units). A little behind, and below the 180 thousand level with 175 thousand, we have Distributor.

Quantity sold by Customer Segment

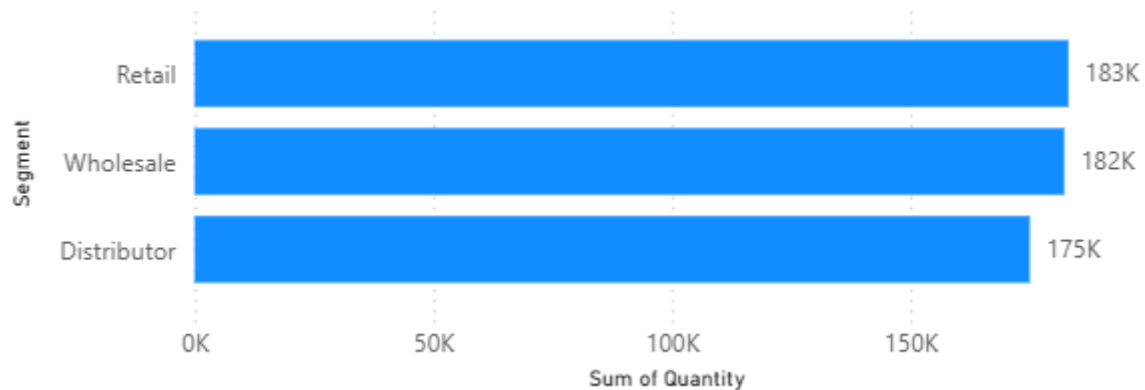


Figure 13 - Quantity sold for each customer's segment

Analysing the profits by segment, it's easy to conclude from the pie chart below that the weights of every segment in the total of profits are very similar, with first place for the Wholesale, then retail and finally distributor.

Profit by Customer's Segment

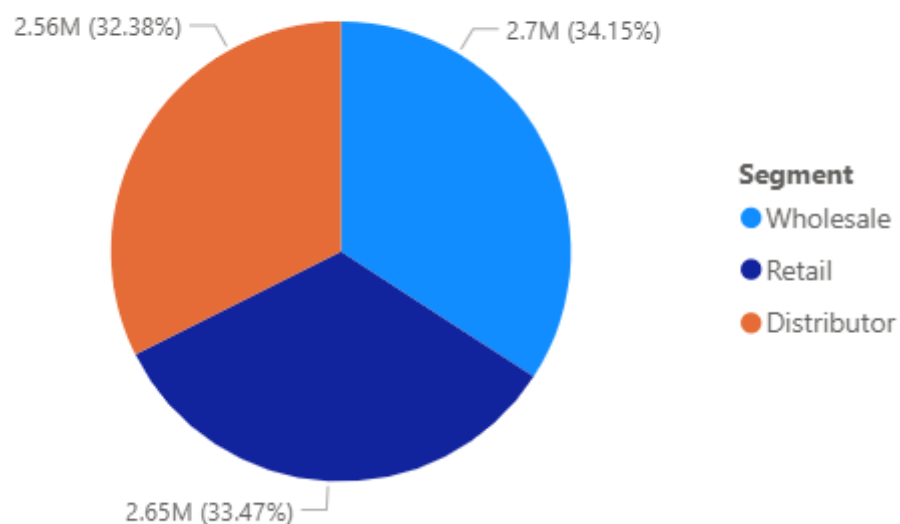


Figure 14 - Distribution of Profit by customer's segment

Making a comparative analysis with the bar chart about the quantity sold we can see that, despite the small difference, more units of our products were sold to Retail, but the profits are higher from the sales to the customers in the Wholesale segment.

Lastly there's a table showing the "declining customers". That means the profits from the sales for those clients in 2024 were lower than in 2023, and in 2023 were lower than in 2022.

To obtain that, a measure was built, from other measures. First, there were created measures to calculate the quantities sold and the profits obtained in each year (2022, 2023 and 2024). Then measures to calculate the declining of quantity sold and profit obtained were calculated. Finally, a last measure was obtained to calculate the declining customers, that follow the criteria previously presented.

Conclusions and recommendations

Despite the increase in revenues and profits through the time frame analysed, work needs to be done in order to improve the company's performance.

Geographic analysis shows that sales are heavily concentrated in Australia. Despite similar customer numbers in the United States (Figure 11), revenue from this market remains significantly lower.

Particular attention should be given to expanding sales in India and the USA, which have significant growth potential. Notably, the United States has a similar number of customers to Australia (Figure 11) but generates much lower revenue.

The other big challenge is Asia. Although being present only in one country, India is currently the most populous country in the world, and many Australian companies are making excellent businesses with India, due to the strong economic and trade relations between the two countries.

Future market expansion should prioritize Asia, given its geographic proximity to Australia, massive consumer base, and strong trade relations.

From the analysis of the products, it is important to understand why the revenues from the brands Velvet Cocoa and SweetPeak are so low compared to the other two brands.

And, as it was explained before, it is relevant for the company to understand how to extrapolate the success they have in Australia to other markets, especially in the neighbouring New Zealand.